

Лабораторная работа №6

Моделирование сетей передачи данных

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Основной целью работы является знакомство с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получение навыков моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.

1. Задайте топологию, состоящую из двух хостов и двух коммутаторов с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по ограничению пропускной способности сети с помощью TBF в эмулируемой глобальной сети.
3. Самостоятельно реализуйте воспроизводимые эксперимент по применению TBF для ограничения пропускной способности. Постройте соответствующие графики.

Выполнение лабораторной работы

<pre>root@mininet-vm:/home/mininet# ifconfig h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500 inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.2 55 ether 2e:43:2b:6b:73:48 txqueuelen 1000 (Ethernet) RX packets 0 bytes 0 (0.0 B) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 0 bytes 0 (0.0 B) TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0 lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536 inet 127.0.0.1 netmask 255.0.0.0 loop txqueuelen 1000 (Local Loopback) RX packets 985 bytes 288508 (288.5 KB) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 985 bytes 288508 (288.5 KB) TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0 root@mininet-vm:/home/mininet#</pre>	<pre>root@mininet-vm:/home/mininet# ifconfig eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500 inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255 ether 08:00:27:06:0f:9f txqueuelen 1000 (Ethernet) RX packets 7272 bytes 1861686 (1.8 MB) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 6035 bytes 4037456 (4.0 MB) TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0 eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500 inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255 ether 08:00:27:76:68:42 txqueuelen 1000 (Ethernet) RX packets 354 bytes 33094 (33.0 KB) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 379 bytes 34239 (34.2 KB) TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0 lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536 inet 127.0.0.1 netmask 255.0.0.0 loop txqueuelen 1000 (Local Loopback) RX packets 19838 bytes 5397058 (5.3 MB) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 19838 bytes 5397058 (5.3 MB)</pre>
<pre>root@mininet-vm:/home/mininet# ifconfig h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500 inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.2 55 ether ea:84:14:ba:38:3f txqueuelen 1000 (Ethernet) RX packets 0 bytes 0 (0.0 B) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 0 bytes 0 (0.0 B) TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0 lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536 inet 127.0.0.1 netmask 255.0.0.0 loop txqueuelen 1000 (Local Loopback) RX packets 1015 bytes 256292 (256.2 KB) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 1015 bytes 256292 (256.2 KB) TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0 root@mininet-vm:/home/mininet#</pre>	<pre>root@mininet-vm:/home/mininet# ifconfig eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500 inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255 ether 08:00:27:06:0f:9f txqueuelen 1000 (Ethernet) RX packets 7329 bytes 1870500 (1.8 MB) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 6100 bytes 4067150 (4.0 MB) TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0 eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500 inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255 ether 08:00:27:76:68:42 txqueuelen 1000 (Ethernet) RX packets 355 bytes 33184 (33.1 KB) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 380 bytes 34329 (34.3 KB) TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0 lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536 inet 127.0.0.1 netmask 255.0.0.0 loop txqueuelen 1000 (Local Loopback) RX packets 19975 bytes 5431706 (5.4 MB) RX errors 0 dropped 0 overruns 0 frame 0 TX packets 19975 bytes 5431706 (5.4 MB)</pre>

Рисунок 1: ifconfig

Выполнение лабораторной работы

```
host: h1*@mininet-vm
TX errors 0 dropped 0 overruns 0 carrier 0 collisions
0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 985 bytes 288508 (288.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 985 bytes 288508 (288.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions
0
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 4
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=6.31 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.362 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.094 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.084 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3041ms
rtt min/avg/max/mdev = 0.084/1.712/6.309/2.656 ms
root@mininet-vm:/home/mininet#

host: h2*@mininet-vm
TX errors 0 dropped 0 overruns 0 carrier 0 collisions
0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1015 bytes 256292 (256.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1015 bytes 256292 (256.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions
0
root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 4
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=3.15 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.101 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.123 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.116 ms

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3035ms
rtt min/avg/max/mdev = 0.101/0.871/3.147/1.313 ms
root@mininet-vm:/home/mininet#
```

Выполнение лабораторной работы

```
host: h1" @mininet-vm
MBytes
[ 7] 2.01-3.01 sec 872 MBytes 7.36 Gbits/sec 0 8.28
MBytes
[ 7] 3.01-4.01 sec 916 MBytes 7.69 Gbits/sec 0 8.28
MBytes
[ 7] 4.01-5.01 sec 916 MBytes 7.68 Gbits/sec 0 8.28
MBytes
[ 7] 5.01-6.00 sec 906 MBytes 7.63 Gbits/sec 0 8.28
MBytes
[ 7] 6.00-7.01 sec 922 MBytes 7.71 Gbits/sec 0 8.28
MBytes
[ 7] 7.01-8.01 sec 936 MBytes 7.85 Gbits/sec 0 8.28
MBytes
^C[ 7] 8.01-8.19 sec 148 MBytes 6.93 Gbits/sec 0 8.
28 MBytes
- - - - -
[ ID] Interval      Transfer      Bitrate      Retr
[ 7] 0.00-8.19 sec 7.13 GBytes 7.48 Gbits/sec 0
sender
[ 7] 0.00-8.19 sec 0.00 Bytes 0.00 bits/sec
receiver
iperf3: interrupt - the client has terminated
root@mininet-vm:/home/mininet#
```

```
host: h2" @mininet-vm
[ 7] 2.00-3.00 sec 883 MBytes 7.41 Gbits/sec
[ 7] 3.00-4.00 sec 917 MBytes 7.67 Gbits/sec
[ 7] 4.00-5.00 sec 915 MBytes 7.70 Gbits/sec
[ 7] 5.00-6.00 sec 907 MBytes 7.60 Gbits/sec
[ 7] 6.00-7.00 sec 922 MBytes 7.75 Gbits/sec
[ 7] 7.00-8.00 sec 936 MBytes 7.83 Gbits/sec
[ 7] 7.00-8.00 sec 936 MBytes 7.83 Gbits/sec
- - - - -
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-8.00 sec 7.13 GBytes 7.66 Gbits/sec
receiver
iperf3: the client has terminated
-----
Server listening on 5201
-----
[ 7]
```

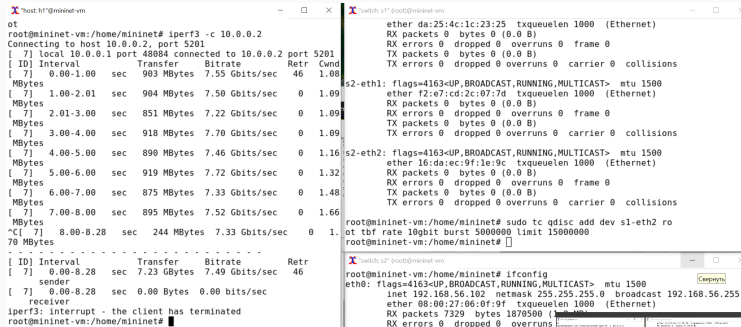
```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root tbf rate 10gbit burst 5000000 limit 15000000
```

Рисунок 4: Ограничение скорости на конечных хостах

Выполнение лабораторной работы

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 48080 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate      Retr  Cwnd
[ 7]  0.00-1.00      sec    883 MBytes  7.40 Gbits/sec    0   4.69
MBytes
[ 7]  1.00-2.01      sec    844 MBytes  6.99 Gbits/sec    0   8.02
MBytes
[ 7]  2.01-3.00      sec    885 MBytes  7.50 Gbits/sec    0   8.02
MBytes
[ 7]  3.00-4.00      sec    891 MBytes  7.48 Gbits/sec    0   8.02
MBytes
[ 7]  4.00-5.00      sec    852 MBytes  7.16 Gbits/sec    0   8.02
MBytes
[ 7]  5.00-6.00      sec    855 MBytes  7.17 Gbits/sec    0   8.02
MBytes
[ 7]  6.00-7.00      sec    870 MBytes  7.29 Gbits/sec    0   8.02
MBytes
[ 7]  7.00-8.00      sec    911 MBytes  7.65 Gbits/sec    0   8.02
MBytes
[ 7]  8.00-9.00      sec    908 MBytes  7.61 Gbits/sec    0   8.02
MBytes
^C[ 7]  9.00-9.83      sec    767 MBytes  7.78 Gbits/sec    0   8.
02 MBytes
- - - - -
[ ID] Interval           Transfer     Bitrate      Retr
[ 7]  0.00-9.83      sec    8.46 GBytes  7.40 Gbits/sec    0
sender
[ 7]  0.00-9.83      sec    0.00 Bytes    0.00 bits/sec
receiver
iperf3: interrupt - the client has terminated
root@mininet-vm:/home/mininet#
```

Выполнение лабораторной работы



The image shows two terminal windows from a Mininet virtual network environment.

The left terminal window shows the output of the `iperf3` command. It displays a table of performance metrics over 10-second intervals. The final summary shows a transfer of 7.23 GBytes at a rate of 7.49 Gbits/sec.

ID	Interval	Transfer	Bitrate	Retr	Cwnd
[7]	0.00-1.00 sec	903 MBytes	7.55 Gbits/sec	46	1.08
[7]	1.00-2.01 sec	904 MBytes	7.50 Gbits/sec	0	1.09
[7]	2.01-3.00 sec	851 MBytes	7.22 Gbits/sec	0	1.09
[7]	3.00-4.00 sec	918 MBytes	7.70 Gbits/sec	0	1.09
[7]	4.00-5.00 sec	890 MBytes	7.46 Gbits/sec	0	1.16
[7]	5.00-6.00 sec	919 MBytes	7.72 Gbits/sec	0	1.32
[7]	6.00-7.00 sec	875 MBytes	7.33 Gbits/sec	0	1.48
[7]	7.00-8.00 sec	895 MBytes	7.52 Gbits/sec	0	1.66
[7]	8.00-8.28 sec	244 MBytes	7.33 Gbits/sec	0	1.70
[ID]	Interval	Transfer	Bitrate	Retr	
[7]	0.00-8.28 sec	7.23 GBytes	7.49 Gbits/sec	46	
[7]	0.00-8.28 sec	0.00 Bytes	0.00 bits/sec		

The right terminal window shows the configuration of two switches, `s2-eth1` and `s2-eth2`. Both interfaces are configured with a maximum transmission unit (MTU) of 1500 and are connected to the `root@mininet-vm`. The output of the `ifconfig` command for `eth0` shows a broadcast address of 192.168.56.255 and a netmask of 255.255.255.0.

Рисунок 6: Ограничение скорости на коммутаторах

Выполнение лабораторной работы

```
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 4
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=14.2 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=11.7 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=13.1 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=10.5 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3067ms
rtt min/avg/max/mdev = 10.546/12.387/14.190/1.384 ms
root@mininet-vm:/home/mininet#

RX packets 0  bytes 0 (0.0 B)
RX errors 0  dropped 0  overruns 0  frame 0
TX packets 0  bytes 0 (0.0 B)
TX errors 0  dropped 0  overruns 0  carrier 0  collisions

root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root
ot tbf rate 10gbit burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# sudo tc qdisc del dev s1-eth2 root
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root handle 1: n
etm delay 10ms
root@mininet-vm:/home/mininet#
```

Рисунок 7: Объединение NETEM и TBF

Выполнение лабораторной работы

```
host: h11@mininet-vm
rtt min/avg/max/mdev = 10.546/12.387/14.190/1.384 ms
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 48088 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr    Cwnd
[ 7]  0.00-1.00    sec      180 MBytes  1.51 Gbits/sec  827    2.53
MBytes
[ 7]  1.00-2.00    sec      221 MBytes  1.86 Gbits/sec    0    2.73
MBytes
[ 7]  2.00-3.00    sec      228 MBytes  1.91 Gbits/sec    0    2.90
MBytes
[ 7]  3.00-4.00    sec      178 MBytes  1.49 Gbits/sec  135    2.15
MBytes
[ 7]  4.00-5.00    sec      189 MBytes  1.58 Gbits/sec    0    2.25
MBytes
[ 7]  5.00-6.00    sec      132 MBytes  1.11 Gbits/sec   90    1.68
MBytes
[ 7]  6.00-7.01    sec      142 MBytes  1.19 Gbits/sec    0    1.76
MBytes
[ 7]  7.01-8.00    sec      152 MBytes  1.29 Gbits/sec    0    1.82
MBytes
[ 7]  8.00-9.00    sec      155 MBytes  1.30 Gbits/sec    0    1.87
MBytes
[ 7]  9.00-10.00   sec      161 MBytes  1.35 Gbits/sec    0    1.89
MBytes
- - - - -
[ ID] Interval      Transfer    Bitrate    Retr
[ 7]  0.00-10.00   sec      1.70 GBytes  1.46 Gbits/sec  1052
sender
[ 7]  0.00-10.02   sec      1.69 GBytes  1.45 Gbits/sec
receiver

switch: s11 (root@mininet-vm)
s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether f2:e7:cd:2c:07:7d txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions
s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether 16:da:ec:9f:1e:9c txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root
ot tbf rate 10gbit burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# sudo tc qdisc del dev s1-eth2 root
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root handle 1: n
etem delay 10ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 parent 1: handle
2: tbf rate 2gbit burst 1000000 limit 2000000
root@mininet-vm:/home/mininet#
```

Рисунок 8: Объединение NETEM и TBF

Выполнение лабораторной работы

```
mininet@mininet-vm: ~/work/lab_tbf
GNU nano 4.8                                lab netem i.py
#!/usr/bin/env python

"""
Simple experiment.
Output: ping.dat
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI

from mininet.log import setLogLevel, info
import time

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s1 = net.addSwitch( 's1' )

    info( '*** Creating links\n' )
    net.addLink( h1, s1 )
    net.addLink( h2, s1 )

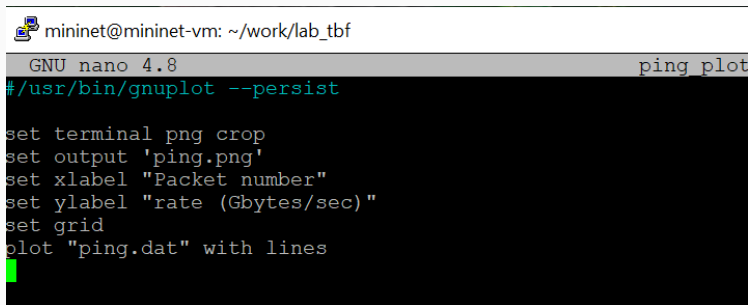
    info( '*** Starting network\n' )
    net.start()

    info( '*** Set rate\n' )
    h1.cmdPrint( 'tc qdisc add dev h1-eth0 root tbf rate 10gbit burst 5000000 limit 15000000' )
    time.sleep(10) # Wait 10 seconds

    info( '*** Starting server on h2\n' )
    h2.cmdPrint( 'iperf3 -s &' )
    info( '*** Running iperf3 client from h1 to h2\n' )
    h1.cmdPrint( 'iperf3 -c ' + h2.IP() + ' | grep "MBytes" | awk \'{print $7}\'} > ping.dat' )

    info( '*** Stopping network' )
    net.stop()

if __name__ == '__main__':
    setLogLevel( 'info' )
    emptyNet()
```

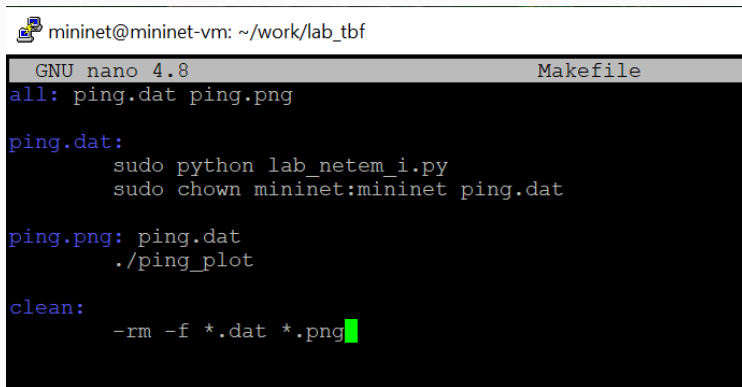


```
mininet@mininet-vm: ~/work/lab_tbf
GNU nano 4.8                                ping_plot
#!/usr/bin/gnuplot --persist

set terminal png crop
set output 'ping.png'
set xlabel "Packet number"
set ylabel "rate (Gbytes/sec)"
set grid
plot "ping.dat" with lines
```

Рисунок 10: Скрипт для отрисовки графика

Выполнение лабораторной работы



The screenshot shows a terminal window with a title bar indicating the user is 'mininet' on a 'mininet-vm' machine, located in the directory '~/work/lab_tbf'. The terminal content shows the GNU nano 4.8 text editor editing a file named 'Makefile'. The Makefile contains three targets: 'all', 'ping.dat', and 'clean'. The 'all' target depends on 'ping.dat' and 'ping.png'. The 'ping.dat' target runs 'sudo python lab_netem_i.py' and 'sudo chown mininet:mininet ping.dat'. The 'ping.png' target depends on 'ping.dat' and runs './ping_plot'. The 'clean' target runs '-rm -f *.dat *.png'.

```
mininet@mininet-vm: ~/work/lab_tbf
GNU nano 4.8 Makefile
all: ping.dat ping.png

ping.dat:
    sudo python lab_netem_i.py
    sudo chown mininet:mininet ping.dat

ping.png: ping.dat
    ./ping_plot

clean:
    -rm -f *.dat *.png
```

Рисунок 11: Makefile

Выполнение лабораторной работы

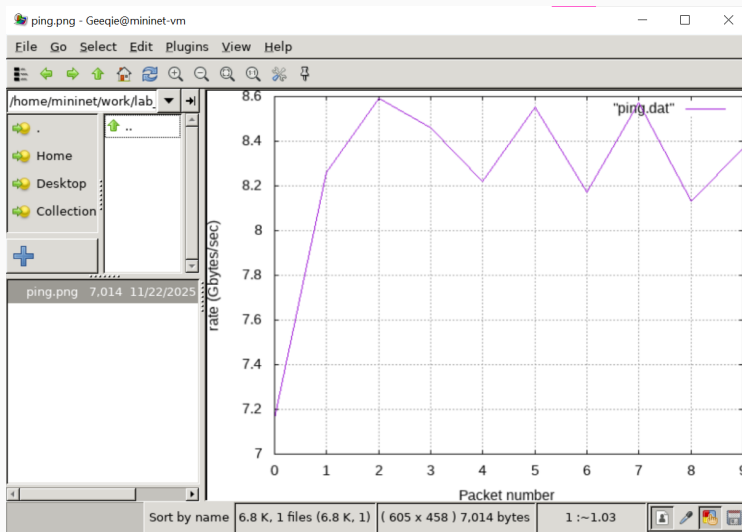


Рисунок 12: График изменения скорости передачи

В ходе выполнения лабораторной работы я познакомилась с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получила навыки моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.